

COMP 202: DATA STRUCTURES AND ALGORITHMS

(Fall 2025 Syllabus)

Instructor: Prof. Dr. Alptekin Küpçü

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Office Hours: Check course calendar on the course web page.

Webpage: courses.ku.edu.tr/comp202

E-mail: Always send e-mail to comp202ta-group@ku for course-related questions. **All other emails will be ignored!**

Prerequisite: Discrete Mathematics and Java. COMP 106 and COMP 132 need to be finished before registering to this course.

Topics:

- Algorithms, Complexity, Recursion.
- Linked Lists, Stacks, Queues.
- Trees, Tries, Skip Lists, Heaps, Hash Tables.
- Graphs and Graph Algorithms.
- Sorting Algorithms.

Objectives:

1. To teach about fundamental data structures: lists, stacks, queues, trees, balanced binary search trees, priority queues, hash tables, graphs.
2. To introduce you to basic searching, sorting, and graph algorithms.
3. To provide a basic understanding of how and where different data structures are employed.
4. To enable you to implement different data structures.

Learning Outcomes: At the end of this course, you will be able to

1. determine appropriate data structures and algorithms to solve problems efficiently,
2. employ fundamental data structures in your algorithms,
3. implement yourself and use others' implementations of widely-employed data structures,
4. analyze time and space complexity of algorithms.

Text Book: Goodrich & Tamassia & Goldwasser, "Data Structures and Algorithms in Java", Wiley. ISBN: 978-0470398807 (required)
Suggested exercises are available for the "international student version" for textbook editions 5 or 6.

Grading: Assignments (20%), Midterm Exam (40%), Final Exam (40%).

- You have to obtain at least 40/100 out of each exam to pass the course.
- Catalog-like system, no curve grading. This means, others' performance does not affect your grade.
Sample grading (may change, provided as an example):

- * 100+: A+
- * 93-100: A
- * 86-92: A-
- * 81-85: B+
- * 75-80: B
- * 70-74: B-
- * 64-69: C+
- * 61-63: C
- * 58-60: C-
- * 55-57: D+
- * 51-54: D
- * 0-50: F

- Grading objection must be done within the announced time period. No online objection or objection on behalf of someone else is allowed.
- If you have an approved health report for midterm exam, your final exam grade will be counted also as your midterm exam grade.
- If you have an approved health report for the final exam, please apply for the make-up exam.
- You have to request makeup by indicating your approved health report, by sending an e-mail to the course e-mail address. There are no automated makeups.
- There is no remedial exam.

Code of Conduct:

Teaching Style: There are pre-recorded videos and textbook readings assigned to you weekly on KUHUB Learn. Follow them *before* attending the lectures to enhance your understanding.

Attendance: Attendance is online through KUHUB Learn. Due to national (YÖK) rules, at least 70% attendance must be marked to pass the course. Please do not step in and out of the classroom when the lecture/lab/PS is in progress (in important cases, ask for my permission *ahead of time*). TAs have the same authority as the instructor, and hence you must follow their rules as well.

Device Use: Use of laptops, tablets, cell phones, or similar devices is **strictly forbidden** in the classroom.

Assignments: **Late assignments will not be accepted unless you have my permission well ahead of time** with a valid excuse. Do not ask for an extension when it is close to the deadline; such requests will not be granted. Hand-written solutions will be accepted as long as they are legible. It is our right to cut off grade points if we cannot read your answer. Always target *the optimal answer* for full grade.

Mandatory: For each assignment, you must sign the following (otherwise your assignment will not be graded):

“I hereby declare that I have completed this individually, without support from anyone else. I hereby accept that only the below listed sources are approved to be used:

- (i) Course textbook,

(ii) All material that is made available to me for this course (on course web page or KUHUB Learn),

(iii) Notes taken by me during the lectures.

I have not used, accessed or taken any unpermitted information from any other source. Hence, all effort belongs to me.”

ID Number, Name Surname, Date, Signature (all written with blue pen).

Cheating: Do not cheat. We employ **cheating detection tools** that can easily detect cheating. By national and university-wide rules, cheating may be penalized with **one or two semesters suspension** from the university. The Koç University Student Code of Conduct includes, but is not limited to, the following: Your work needs to be your own. It cannot be joint work with any other person (in the class or not). You cannot have someone do the assignment for you (paid or not). You cannot submit an assignment that is based on submissions in earlier semesters. You cannot use a solution manual, or an online solution to the question. To get a grade (other than F) in this course, you must fully comply with the Koç University Student Code of Conduct. Attending any Koç University course means you have read and agreed to this conduct, the course syllabus, and honor codes. Failure to comply will result in **a direct F for your letter grade for the course and disciplinary action**.

Exam/Quiz Rules: The examination is individual, and no collaboration is allowed. The examination is closed book, closed notes. No physical or digital equipment (including but not limited to cell phones, tablets, smart watches, smart glasses, headphones, cameras, microphones, calculators) is permitted. Put all equipment in your bags or coat pockets. Never leave them visible. Never use them during the examination. At the end of the examination, you must return all the examination sheets.

Acting contrary to these regulations, or not handing over physical or digital equipment even upon detection, or not following the instructions of the exam proctors will be considered as cheating or attempted cheating; your examination will be terminated, an official report will be issued, and a disciplinary investigation will be initiated.

AI Usage: Artificial Intelligence (AI) systems must *not* be used for graded assignments in this course. Koç University Office of Learning and Teaching (KOLT) and Information Technology (IT) details AI usage rules:

kolt.ku.edu.tr/artificial-intelligence/
it.ku.edu.tr/en/use-of-generative-ai/

Students must read, understand, and adhere to these rules and policies. It is the student's responsibility to check the accuracy of the information generated by AI. Direct copying and pasting from AI tools is strictly prohibited. The use of AI must be indicated and properly cited, when explicitly allowed.

How to Succeed:

1. You need to do **lots of reading**. Do not expect to understand hard topics without reading from *multiple* sources.
2. You need to do **lots of programming**. Do not expect to succeed without *implementing fully-working* programs yourself.
3. Make sure to do the assignments and suggested exercises yourself, without finding the solution somewhere else.
4. You need to solve **more exercises than the ones assigned**. The assignments will help, but they will **not** be enough to guarantee success.

5. If you did not understand a particular topic, **ask** before we switch to the next topic. Your learning will improve if you join in-class discussions. **Work regularly.**
6. Failure to notice an announcement is not an excuse. It is your responsibility to keep up-to-date with the course material.
7. Do not forget that this is one of the most fundamental and important courses for Computer Engineering. Try to learn the topics in the best possible way.
8. **Do not cheat.**